

AI Experiment Consultant

From an idea, pain point, or screenshot to an implementation-ready UpGrade experiment plan

PELE 2026 · Work-in-Progress / Demo

Zack Lee and April Murphy

Carnegie Learning

The onboarding problem

- **UpGrade** is Carnegie Learning's open-source platform for educational A/B testing
- It supports **running and managing** experiments — *once a team has a clear experiment plan*
- The hard part often comes **earlier**: turning a rough idea, pain point, or specific interaction into that plan
- *What* should we test? *Where* does condition assignment happen? *Which* conditions and metrics? *What needs to change in the app?*
- Today, that planning gap often requires **expert consultation**

What it does

- A web-based, **chat-driven** consultant for educational software teams
- **Input:** an idea, a pain point, or a screenshot
- **Output:** a concrete A/B test plan + an implementation-ready **markdown report** tailored to UpGrade
- **Human-controlled** and **planning-focused** — the tool suggests, the user decides

Six-phase consulting workflow

1. Learning app description
2. Page / problem / interaction description
3. Ideation & hypothesis refinement (*optional research grounding*)
4. UpGrade experiment design
5. Synthetic preflight simulation (*optional*)
6. Report generation

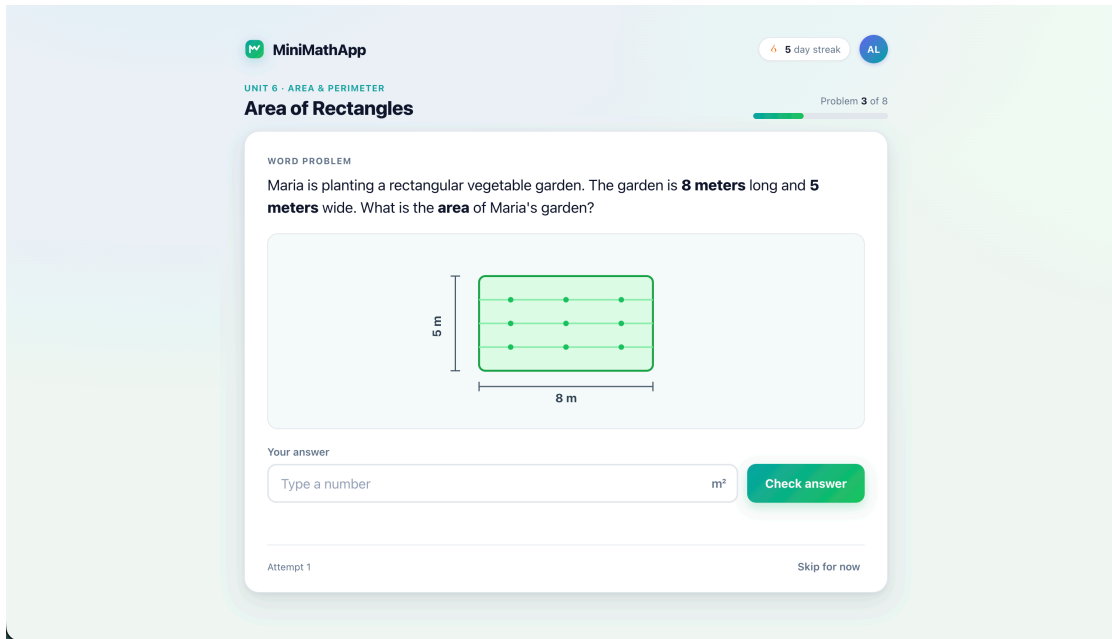
The user approves every major transition.

Why the report matters

- The conversation ends as a **structured markdown report**
- It captures the **hypothesis, UpGrade experiment design, simulation summary, and implementation guidance**
- It gives researchers, developers, and product teams **one shared artifact**
- It can also serve as a **concrete spec** for later implementation work

Live demo — MiniMathApp

- A **fictional** math-practice app · an area word problem (a rectangular garden)
- Pain point: students often **get stuck or answer incorrectly on the first try**



Next: we give the consultant this screenshot and ask it to help plan an experiment →

Scope today, future direction

- Today: a **planning-focused MVP** for simple, concrete UpGrade experiment designs
- Synthetic preflight shows **UpGrade mechanics**, not evidence of learning effects
- Not yet evaluated with real teams — planned during **Fall 2026 onboarding**
- Future: an approved report could connect planning, implementation, UpGrade setup, and analysis in a **human-reviewed pipeline**

Thank you

Questions?

Try the app: <https://upgrade-demo.carnegielearning.com/ai-consultant>

Repository: <https://github.com/CarnegieLearningWeb/ai-experiment-consultant>

Contact: zlee@carnegielearning.com